

Confusion Matrix

A Complete Guide to Classification Evaluation

The **Confusion Matrix** is a foundational tool for evaluating classification models. Rather than collapsing performance into a single number, it breaks down every prediction the model makes into four categories — showing not just how often the model is right, but precisely *how* and *where* it goes wrong.

Cell	Full Name	Meaning	Impact On
TP	True Positive	Correct Positive prediction	Precision, Recall
TN	True Negative	Correct Negative prediction	Accuracy, Specificity
FP	False Positive	Wrong — said Positive, was Negative	Precision ↓
FN	False Negative	Wrong — said Negative, was Positive	Recall ↓

Table — The four cells at a glance

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1. What Is a Confusion Matrix?

A Confusion Matrix is a **N×N table** (N = number of classes) that summarises the performance of a classification model by comparing predicted labels against actual (true) labels. For binary classification — the most common case — it is a **2×2 grid** with rows representing actual classes and columns representing predicted classes.

The Binary Confusion Matrix

	Predicted: POSITIVE	Predicted: NEGATIVE
Actual: POSITIVE	TRUE POSITIVE (TP) Model correctly predicted Positive	FALSE NEGATIVE (FN) Model missed a real Positive
Actual: NEGATIVE	FALSE POSITIVE (FP) Model wrongly predicted Positive	TRUE NEGATIVE (TN) Model correctly predicted Negative

Figure 1 — Binary Confusion Matrix (green = correct, red/orange = errors)

2. The Four Cells Explained in Depth

True Positive (TP)

The model predicted Positive — and it was right.

These are the "hits." The model correctly identified a real positive instance. More TPs = better performance. Every metric you care about improves as TP grows.

Example: A spam filter marks an email as spam, and it genuinely is spam.

True Negative (TN)

The model predicted Negative — and it was right.

These are the "correct rejections." The model correctly identified a real negative instance. In imbalanced datasets, TN can be very large without reflecting useful model skill.

Example: A spam filter marks a legitimate email as not-spam, and it genuinely is not.

False Positive (FP)

The model predicted Positive — but it was wrong.

Also called a **Type I Error** or "false alarm." The model raised an alert for something that was actually negative. FPs hurt **Precision** directly.

Example: A spam filter marks a legitimate email as spam. The user loses an important message.

False Negative (FN)

The model predicted Negative — but it was wrong.

Also called a **Type II Error** or "miss." The model failed to detect a real positive. FNs hurt **Recall** directly and are often the costlier error in high-stakes domains.

Example: A cancer screening model clears a patient who actually has cancer.

3. Key Metrics Derived from the Confusion Matrix

The Confusion Matrix is not just a visual aid — every major classification metric is mathematically derived from its four cells. Knowing these derivations lets you understand exactly what each metric measures and what it ignores.

Metric	Formula	Measures	Range
Accuracy	$(TP+TN) / (TP+TN+FP+FN)$	Overall correct predictions	0 – 1
Precision	$TP / (TP+FP)$	Exactness of positive predictions	0 – 1
Recall	$TP / (TP+FN)$	Coverage of actual positives	0 – 1
Specificity	$TN / (TN+FP)$	Coverage of actual negatives	0 – 1
F1 Score	$2 \times P \times R / (P+R)$	Harmonic mean of P & R	0 – 1
MCC	$(TP \times TN - FP \times FN) / \sqrt{(...)}$	Balanced, all four cells used	-1 – 1
FPR	$FP / (FP+TN)$	False alarm rate	0 – 1
FNR	$FN / (FN+TP)$	Miss rate	0 – 1

Table 1 — All major classification metrics and their CM derivations

Deep Dive: Accuracy vs F1

Accuracy is the most intuitive metric but the most dangerous on imbalanced data. Consider a dataset with 990 negatives and 10 positives — a model that always predicts Negative achieves **99% accuracy** while having **0% recall**. The Confusion Matrix immediately exposes this: TP=0, FN=10, TN=990, FP=0.

Model	TP	TN	FP	FN	Accuracy	Precision	Recall	F1
Always Negative	0	990	0	10	99.0%	N/A	0.0%	0.00
Balanced	7	970	20	3	97.7%	25.9%	70.0%	0.38
Good model	9	985	5	1	99.4%	64.3%	90.0%	0.75

Table 2 — Why high accuracy can be completely misleading

Specificity vs Recall

Specificity (True Negative Rate) and Recall (True Positive Rate) are mirror metrics — one measures how well the model handles the negative class, the other the positive class. The **ROC**

curve plots Recall (TPR) on the y-axis against 1–Specificity (FPR) on the x-axis.

Matthews Correlation Coefficient (MCC)

MCC uses all four cells of the confusion matrix and is considered the most informative single metric for binary classification, especially under class imbalance. A value of +1 = perfect predictions, 0 = no better than random, –1 = perfectly wrong.

$$\text{MCC} = (\text{TP} \times \text{TN} - \text{FP} \times \text{FN}) / \sqrt{[(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})]}$$

Key insight: Accuracy ignores the imbalance between classes. Precision and Recall ignore true negatives entirely. MCC is the only metric that accounts for all four cells equally — making it the gold standard for imbalanced classification.

4. Multi-Class Confusion Matrix

When there are more than two classes, the Confusion Matrix expands to $N \times N$. Each row represents an **actual** class and each column a **predicted** class. The diagonal cells (top-left to bottom-right) represent correct predictions; all off-diagonal cells represent errors.

Example: 3-Class Matrix (Cat / Dog / Bird)

Actual \ Predicted	Cat	Dog	Bird
Cat	50	3	2
Dog	4	45	1
Bird	1	2	47

Figure 2 — 3-class Confusion Matrix (green diagonal = correct, red = errors)

Reading a Multi-Class Matrix

- **Diagonal cells** (e.g., Cat→Cat = 50): correctly classified instances for that class.
- **Off-diagonal rows** (e.g., Cat→Dog = 3): actual Cats misclassified as Dogs — these are the model's False Negatives for "Cat".
- **Off-diagonal columns** (e.g., Dog→Cat = 4): actual Dogs predicted as Cats — these are the model's False Positives for "Cat".
- Each class has its own TP, FP, FN, TN computed by treating it as the positive class in a one-vs-rest scheme.

Per-Class Metrics from the 3-Class Example

Class	TP	FP	FN	Precision	Recall	F1
Cat	50	5	5	0.909	0.909	0.909
Dog	45	5	5	0.900	0.900	0.900
Bird	47	3	3	0.940	0.940	0.940

Table 3 — Per-class metrics derived from the multi-class matrix

Averaging Strategies for Multi-Class

Strategy	How Computed	Use When
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Macro	Simple mean of per-class metrics	All classes equally important
Micro	Pool TP/FP/FN globally across classes	Overall instance-level accuracy matters
Weighted	Mean weighted by class sample count	Class imbalance is present

Table 4 — Multi-class metric averaging strategies

Tip: In a multi-class matrix, a strong model has large numbers on the diagonal and near-zero values everywhere else. Patterns in off-diagonal cells reveal which classes the model confuses with each other — invaluable for feature engineering.

5. Worked Example — Step by Step

Let's build a complete Confusion Matrix from scratch and derive every metric from it.

Scenario

A fraud detection model is evaluated on **10 000 transactions**. 200 are genuinely fraudulent (positive class), 9 800 are legitimate (negative class). The model's predictions are recorded below.

- Fraudulent transactions correctly flagged (TP) = **160**
- Fraudulent transactions missed — flagged as legitimate (FN) = **40**
- Legitimate transactions wrongly flagged as fraud (FP) = **80**
- Legitimate transactions correctly cleared (TN) = **9 720**

Step 1 — Fill in the Confusion Matrix

	Predicted: FRAUD	Predicted: LEGITIMATE
Actual: FRAUD	TP = 160	FN = 40
Actual: LEGITIMATE	FP = 80	TN = 9 720

Figure 3 — Completed Confusion Matrix for the fraud scenario

Step 2 — Compute Every Metric

Metric	Calculation	Result	Interpretation
Accuracy	$(160+9720) / 10000$	98.8%	Misleading — skewed by large TN
Precision	$160 / (160+80)$	0.667	1 in 3 fraud alerts is wrong
Recall	$160 / (160+40)$	0.800	Catches 80% of real fraud
Specificity	$9720 / (9720+80)$	0.992	Rarely bothers legit customers
F1 Score	$2 \times 0.667 \times 0.8 / (0.667+0.8)$	0.727	Balanced precision/recall
FPR	$80 / (80+9720)$	0.008	Low false alarm rate
FNR	$40 / (40+160)$	0.200	20% of fraud slips through

Table 5 — All metrics derived from the fraud detection confusion matrix

Step 3 — Interpret & Act

- Accuracy (98.8%) is deceptive — the model would score 98% even if it never detected any fraud.
- Recall of 0.80 means 20% of real fraud (40 cases) slips through — a serious business risk.
- Precision of 0.667 means one third of fraud alerts are false — creating customer friction.
- **Action:** Lower the decision threshold slightly to improve Recall, accepting some Precision loss.

Always read the full matrix, never just accuracy. In this example, a naive "always predict legitimate" model would score 98% accuracy while catching zero fraudulent transactions. The Confusion Matrix makes this failure immediately visible.

6. Interpreting Results & Common Pitfalls

How to Read Any Confusion Matrix

Check the diagonal first	The diagonal (top-left to bottom-right) holds correct predictions. Large diagonal values and small off-diagonal values = good model.
Identify the dominant error type	Are off-diagonal values larger in the FP cell or FN cell? This tells you whether the model over-predicts or under-predicts the positive class.
Consider class frequencies	A large TN cell on an imbalanced dataset is not impressive — it may just reflect the model defaulting to the majority class. Normalise by row to see class-level rates.
Look for systematic confusion	In multi-class matrices, consistent off-diagonal patterns reveal which classes the model confuses. This guides feature engineering and data collection.

Common Pitfalls

Pitfall	Problem	Fix
Trusting accuracy alone	On imbalanced data, accuracy is useless as a primary metric	Use F1, MCC, or AUPRC instead
Ignoring FN cost	In medical/safety contexts, missed positives are catastrophic	Track Recall and FNR explicitly; use F2 score
Ignoring FP cost	In spam filters, blocking legitimate mail damages trust	Track Precision; raise threshold if FP is costly
Not normalising	Raw counts hide per-class error rates in imbalanced sets	Divide each row by its total (row normalisation)
Single threshold evaluation	Evaluating at 0.5 only misses the full performance picture	Plot the full PR curve or ROC curve

Table 6 — Common confusion matrix pitfalls and fixes

Python Code

```

from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay,
    classification_report
import matplotlib.pyplot as plt

y_true = [0,1,1,0,1,1,0,0,1,0]
y_pred = [0,1,0,0,1,1,1,0,1,0]

# Raw confusion matrix
cm = confusion_matrix(y_true, y_pred)
print(cm)

# Visual plot
disp = ConfusionMatrixDisplay(cm, display_labels=["Neg", "Pos"])
disp.plot(cmap="Blues")
plt.title("Confusion Matrix")
plt.show()

# Full report with all derived metrics
print(classification_report(y_true, y_pred))

```

Key Takeaways

- The Confusion Matrix breaks predictions into TP, TN, FP, FN — the source of every metric.
- TP and TN are correct; FP (Type I) and FN (Type II) are the two error types.
- All major metrics — Accuracy, Precision, Recall, F1, MCC — derive from these four cells.
- Multi-class matrices expand to $N \times N$; read the diagonal for correctness, off-diagonal for confusion patterns.
- Never rely on accuracy alone. Use the full matrix to understand where and how your model fails.